OFFICIAL ASSESSMENT SUBMISSION REPORT · 100 MARKS (15% OF MODULE)

Disaster Management LK

A Community-Powered Disaster Reporting & Emergency Situational Awareness Platform for Sri Lanka

ASSIGNED GROUP ID
[Insert Group ID, e.g., WE_04]

DATE OF ASSESSMENT
4th September 2026 (In-Class 4-Hour Build)

GIT REPOSITORY URL
https://github.com/VTN02/Community-Disaster.git

LIVE DEPLOYED APPLICATION
https://community-disaster.vercel.app/

2-MINUTE VIDEO DEMONSTRATION LINK
[Insert OneDrive / YouTube / Google Drive Video Link]

1. TEAM MEMBERS & CONTRIBUTION BREAKDOWN

Student ID	Student Full Name	Role / Focus Area	Key Contributions & Modules Owned
[IT Number 1]	**Vijaya Kumar Vithusan**	Functional Implementation & Backend Architecture	Designed Express REST controllers, MongoDB Mongoose models, JWT authentication, Windows DNS SRV resolver fix for MongoDB Atlas, database seeding scripts, and backend hosting configuration.
[IT Number 2]	**Y. H. Prasad**	UI Development & Component Architecture	Built React components with Tailwind CSS; created executive Hero section with DMC operational visual assets (`dmc_slider_04.jpg`), Navbar, Disaster Cards, and mobile drawer.
[IT Number 3]	**[Member 3 Full Name]**	Forms, Validation & Interactive Maps	Implemented Disaster Report Form using React Hook Form + Zod, Leaflet/OpenStreetMap GPS pin integration, client-side input validation, and user feedback error alerts.
[IT Number 4]	**[Member 4 Full Name]**	Testing, CI/CD, Documentation & Demo	Conducted end-to-end integration testing, managed Git branching and commit logs, configured Vercel production deployment, and coordinated the 2-minute demonstration video.

2. PROBLEM FRAMING: THE SRI LANKAN CONTEXT

2.1 The Problem

Sri Lanka experiences frequent natural disasters, including severe monsoon-induced flash floods along the Kelani, Kalu, and Gin river basins, dangerous landslides across mountainous terrains (Kandy, Badulla, Kegalle, Nuwara Eliya), and coastal storms. During critical emergency hours, citizens face significant life-safety hazards:

- **Information Delays:** Formal disaster bulletins often take hours to cascade through official channels to rural and suburban citizens.
- **Lack of Real-Time Ground Intelligence:** Commuters and families do not know if specific bridges, roads, or town centres are impassable or flooded.
- **Social Media Misinformation:** Rumors without geolocation checks spread quickly, confusing affected families and misallocating relief resources.
- **Fragmented Emergency Contacts:** Ordinary citizens struggle to quickly find direct emergency numbers (DMC, Suwa Seriya, Police, Fire) under pressure.

2.2 Target Beneficiaries

Ordinary citizens across Sri Lanka's 25 districts, daily commuters, vulnerable families in flood/landslide zones, and first responders (1990 Suwa Seriya ambulance crews, Police, and local divisional secretariats).

3. THE PROPOSED SOLUTION: DISASTER MANAGEMENT LK

Disaster Management LK is a community-powered disaster reporting and situational awareness platform built for rapid public use and administrative verification:

- **Zero-Barrier Reporting:** Citizens can report local incidents in under 60 seconds without login or signup barriers.
- **Precision Geolocation:** Automatically captures GPS coordinates via browser Geolocation API or allows manual pin drops on an interactive map.
- **Interactive Disaster Map:** Full-screen Leaflet & OpenStreetMap visualization with color-coded severity markers (Critical, High, Medium, Low).
- **Administrative Verification Pipeline:** Designated coordinators inspect, verify, update response status, and filter out misinformation.
- **Speed-Dial Emergency Hotlines:** Direct tap-to-call links for **117 (DMC)**, **119 (Police)**, **1990 (Suwa Seriya Ambulance)**, and **110 (Fire & Rescue)**.

4. TECHNOLOGY STACK JUSTIFICATION

Layer	Technology Chosen	Justification & Rationale
Frontend SPA	React 18 + Vite 5	Declarative component architecture, instantaneous HMR development, and fast sub-10s production builds.
Styling & Design	Tailwind CSS 3	Utility-first CSS, mobile-first responsive breakpoints, and consistent emergency-grade color tokens with zero runtime overhead.
Form Validation	React Hook Form + Zod	Uncontrolled inputs for performance and strict schema validation providing user-friendly, real-time error messages.
Geospatial Maps	Leaflet + React Leaflet + OSM	Completely free and open-source mapping requiring no proprietary API keys, credit cards, or query quotas.
Backend API	Node.js + Express.js	Non-blocking asynchronous I/O runtime optimal for handling concurrent citizen disaster reports and lightweight JSON APIs.
Database	MongoDB Atlas + Mongoose	Document database accommodating geospatial coordinates and flexible disaster report schemas. Hosted on cloud Atlas replica set.
Authentication	JWT + bcryptjs (12 rounds)	Stateless token-based authorization for administrative triage endpoints and password hashing.
Hosting	Vercel (Edge CDN)	Global low-latency deployment with automated HTTPS and instant cache invalidation.

5. COMPLIANCE WITH 10 MINIMUM SOFTWARE REQUIREMENTS

#	Requirement	Implementation Evidence in Solution	Status
1	Clear landing page / UI	Executive landing page with national trust badges, DMC operational showcase (`dmc_slider_04.jpg`), and live KPI cards.	Completed
2	Explanation of Sri Lankan problem	Dedicated in-app problem descriptions on Home and About pages highlighting floods, landslides, and road blockages across 25 districts.	Completed
3	At least two functional features	1. Interactive Nationwide Disaster Map with severity pins. 2. Real-time Incident Filter & Search Engine. 3. Admin Verification Workflow.	Completed
4	Form accepting user input	Disaster Incident Reporting Form with GPS location picker, dropdown selectors, area, and description inputs.	Completed
5	Input validation & error messages	Zod schema enforcing description length, district selection, and coordinate boundaries with red error text.	Completed
6	Search, filter, calculate & process	Keyword search, multi-filter (type, severity, district, status), and Haversine formula calculation for proximity distance.	Completed
7	Responsive desktop & mobile UI	Tailwind breakpoints (`sm` to `xl`) with mobile slide-out menu and sliding filter drawer.	Completed

#	Requirement	Implementation Evidence in Solution	Status
8	Basic navigation between screens	Sticky navigation bar with active route indicators and footer linking to all pages.	Completed
9	Sample data for chosen problem	12 realistic pre-seeded Sri Lankan disaster incidents across Colombo, Kandy, Gampaha, and Ratnapura, plus 8 verified emergency contacts.	Completed
10	Demonstrated value to Sri Lankan users	Empowers public safety, provides instant one-tap 117/1990 hotlines, and offers pre-disaster safety checklists.	Completed

6. USAGE OF AI — CLEAR FRAMEWORK & PROMPT LOG (MANDATORY)

MANDATORY AI USAGE DECLARATION (ASSIGNMENT SECTION 2.3)

"This project was built with the assistance of **Antigravity AI (Google DeepMind)**. AI was utilized for component architecture, responsive Tailwind styling, Express REST API routing, troubleshooting Windows DNS SRV queries on MongoDB Atlas, and generating realistic Sri Lankan seed data. Every generated file was reviewed, customized, and verified by the development team. All team members understand and take full ownership of the submitted codebase."

6.1 Mandatory AI Prompt Log (Section 2.2)

#	AI Tool Used	Exact Prompt Submitted	Purpose	Output Verification & Modifications
1	**Antigravity AI** (Google DeepMind)	`"can you clone https://github.com/VTN02/Community-Disaster.git to this project"`	Clone team repository into working environment.	Verified repository structure, files, and git history.
2	**Antigravity AI** (Google DeepMind)	`"yes do it"` *(Follow-up on dependency install and environment setup)*	Install npm dependencies and seed MongoDB Atlas database.	Identified Windows Node c-ares DNS SRV error (`querySrv ECONNREFUSED`). Added `dns.setServers(['8.8.8.8', '1.1.1.1'])` to `db.js` and `seedData.js`. Verified successful connection and data seeding.
3	**Antigravity AI** (Google DeepMind)	`"project working well.. now i want to design it more professional .. i also add picture to public folder can you add it to landing page.."`	Upgrade landing page into an executive portal incorporating `dmc_slider_04.jpg`.	Reviewed generated layout, adjusted mobile padding, added top emergency alert ribbon and speed-dial hotline grid. Verified production build with `npm run build` (0 errors).
4	**Antigravity AI** (Google DeepMind)	`"now i want to design this page more professionally https://community-disaster.vercel.app/disasters"`	Upgrade disaster reports and search page to professional standards.	Refactored `FilterPanel.jsx` with quick preset chips; updated `DisasterCard.jsx` to support both Grid and List view modes; verified live client-side sorting and filter tag removals.

7. DEMONSTRATION VIDEO SCRIPT & GUIDE (2 MINUTES)

Timestamp	Section	Talking Points & Visual Demonstration Flow
0:00 - 0:25	Problem & Introduction	Introduce Team [Group ID]. Explain the Sri Lankan crisis: monsoon flash floods and landslides cause life safety threats while official news is delayed. Introduce Disaster Management LK.
0:25 - 0:55	Reporting & Leaflet Map	Demonstrate the **Report Disaster** form: capture GPS location with one click, select hazard type, fill description, and show instant validation. Switch to the **Disaster Map** to show colored severity pins.
0:55 - 1:25	Filtering & Speed Dial	Show the **Active Reports** page: demonstrate keyword search, filter by district (e.g. Colombo, Kandy), switch between Grid and List views, and show one-tap emergency calling (117 / 1990).
1:25 - 1:45	Admin Verification Portal	Log in with `admin@disasterlk.gov.lk`: show the moderation queue, verify a pending report, and show how verified status reflects on the public map.
1:45 - 2:00	Deployment & Impact	Highlight live Vercel deployment (`community-disaster.vercel.app`) and mobile responsiveness. Conclude with community impact statement.

8. ACADEMIC INTEGRITY & FINAL DECLARATION

We hereby certify that this project and accompanying documentation represent original work developed during the scheduled SE3090 Assignment 2 Mini Hackathon session on 4th September 2026. All framework integrations, third-party libraries, and AI assistive tools have been transparently declared in accordance with the SLIIT Student Code of Conduct and the CLEAR framework guidelines.

Team Lead Signature: Vijaya Kumar Vithusan
Date: 4th September 2026

Faculty Evaluation: SLIIT Faculty of Computing
Module: SE3090 Software Engineering Frameworks